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Embedded Systems

3-Wheel Balancing Robot Write-Up

Introduction:

In this project, I intended to design a self-balancing 2-wheel robot. In the end, I ended up creating a 3-wheel robot that is able to be controlled from a separate wireless device such as a cellphone. To accomplish this, I used a code block diagram, KiCad for PCB design, soldering, and STM32Cube programming.

Design Stage:

To start, I used a block diagram to express the system logic behind the self-balancing robot. Here is the initial block diagram and the final (used) block diagram:

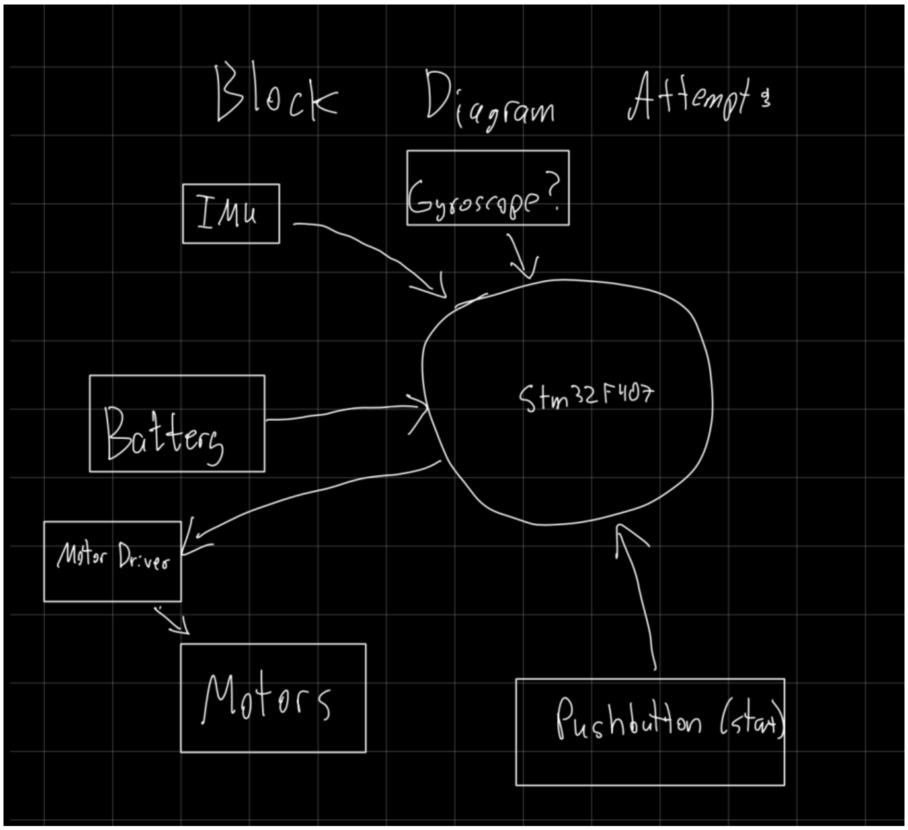

Figure 1: First Attempted System Block Diagram

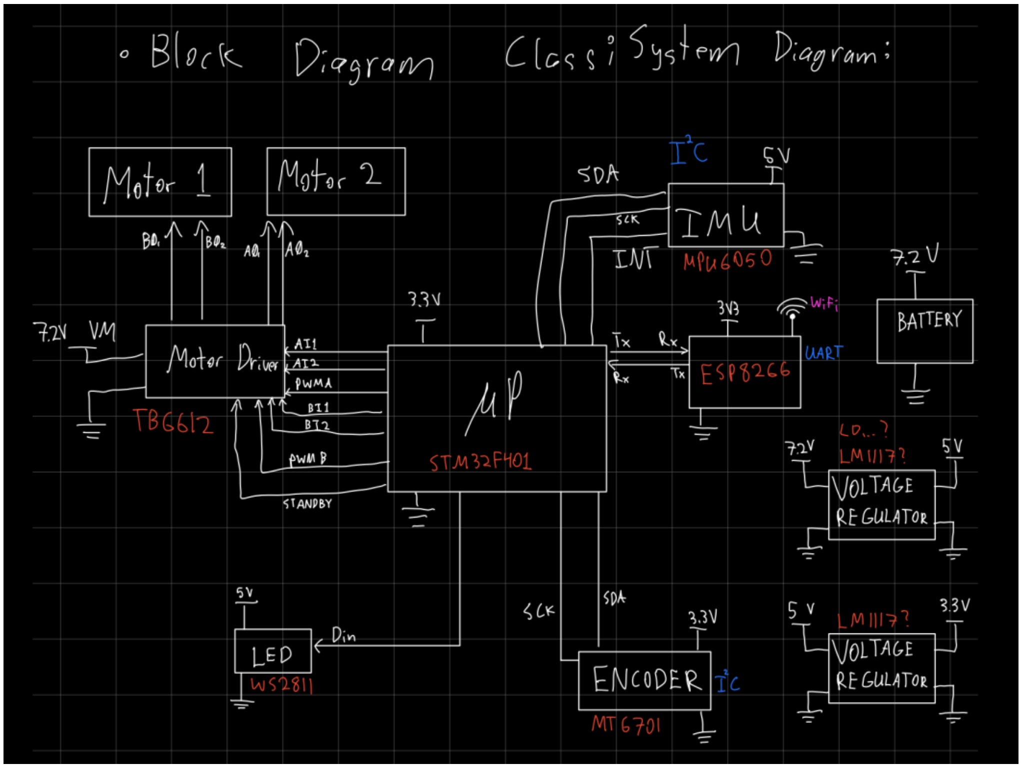

Figure 2: Finalized System Block Diagram

After finalizing the system block diagram and selecting the microprocessor (STM32F401), I was able to move on to PCB design.

PCB Design:

Using KiCad, the following initial schematic was created:

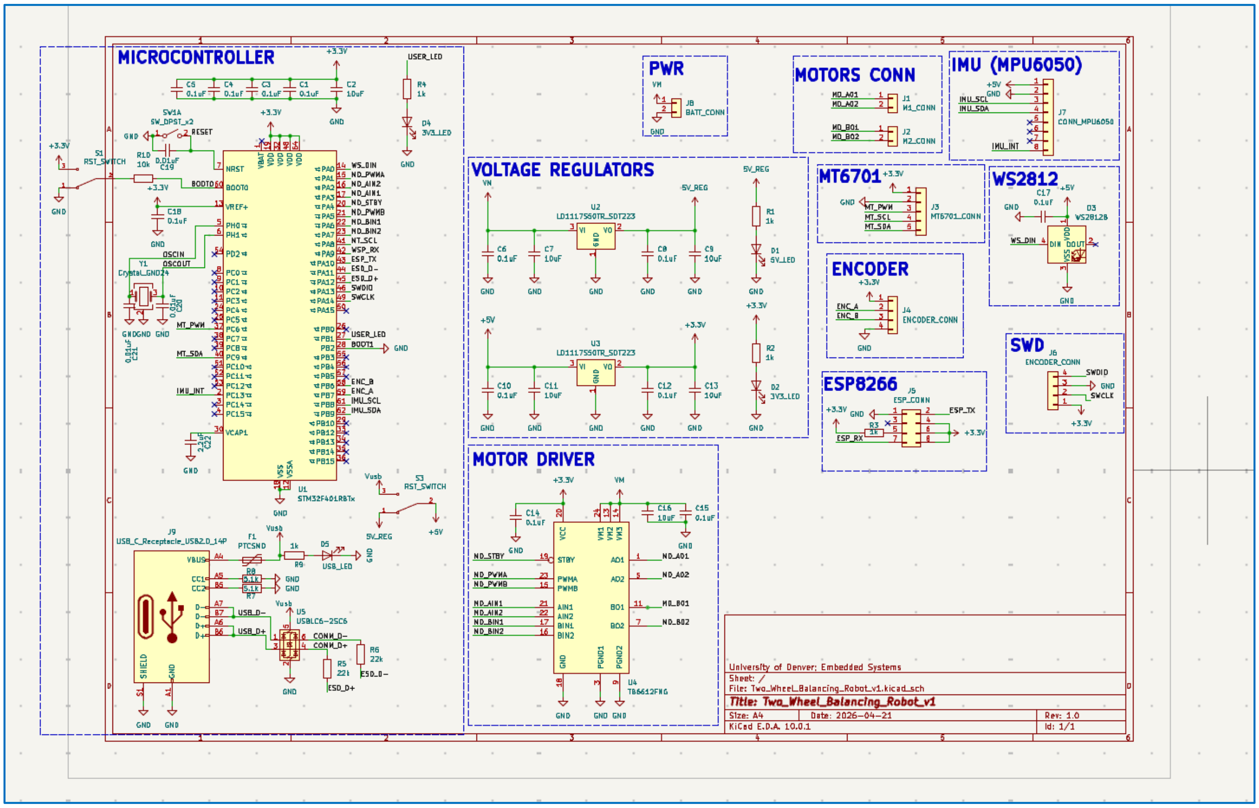

Figure 3: Initial System Schematic

After creating my schematic, I had some missing libraries for components that I needed for my circuit, so I used the class schematic:

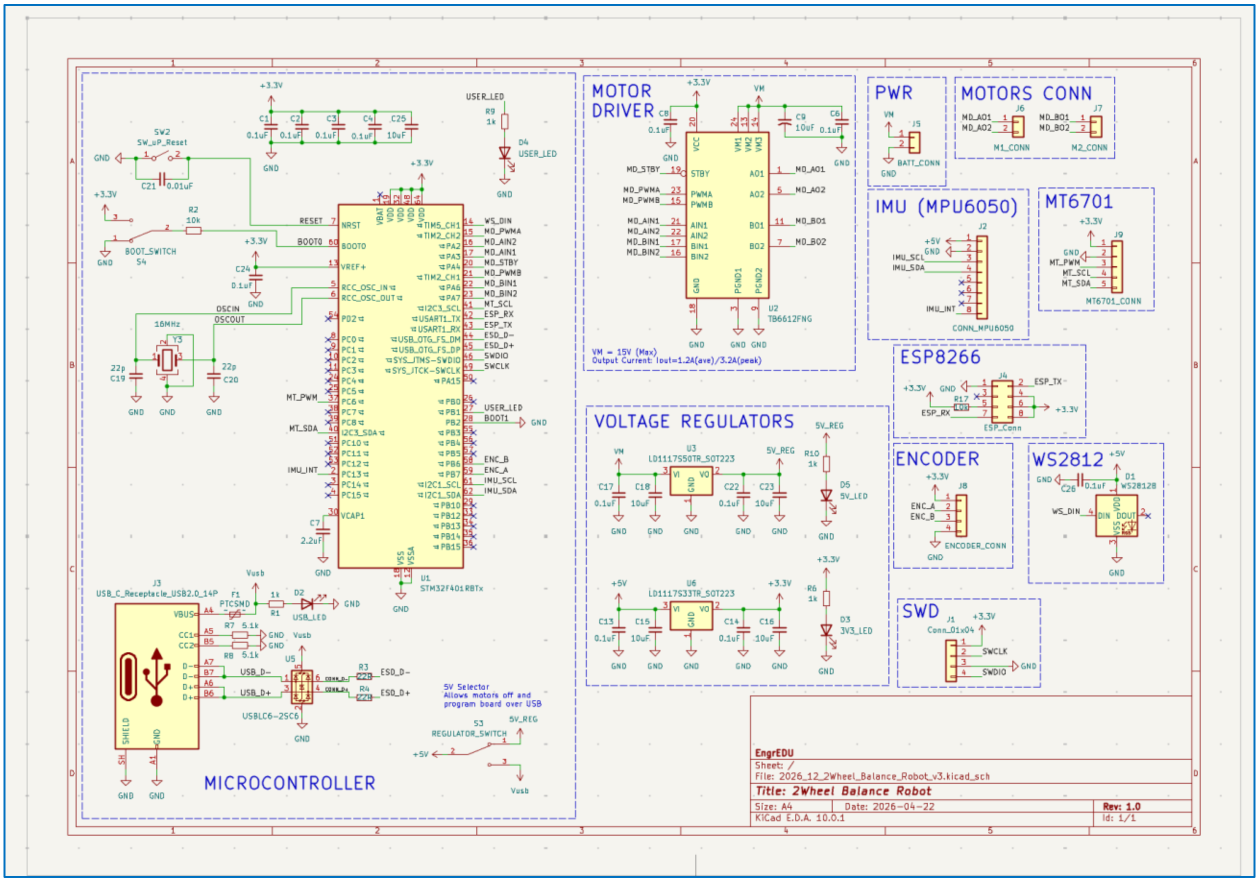

Figure 4: Class System Schematic (Used)

Next, I created my PCB layout:

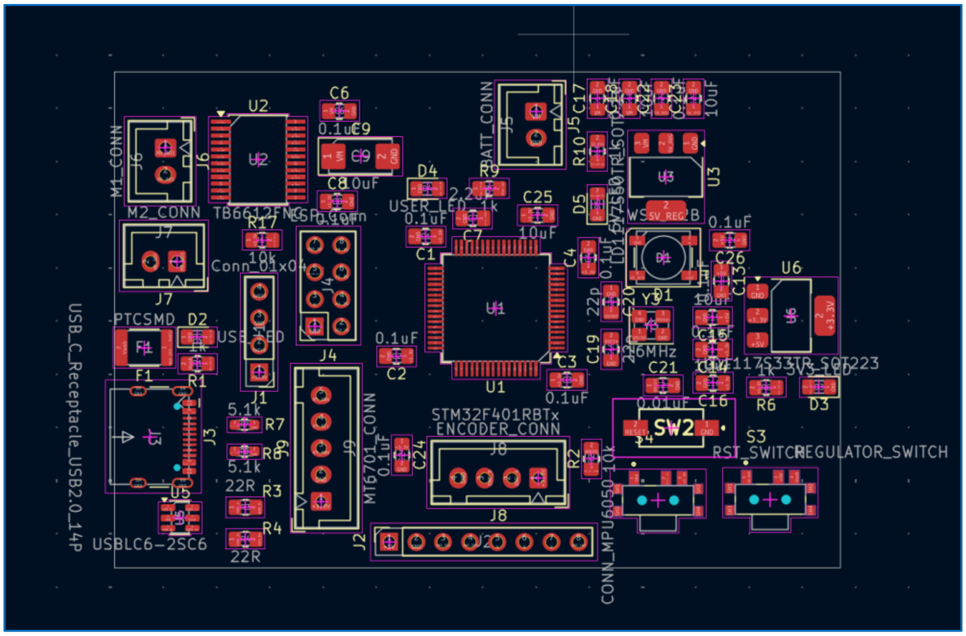

Figure 5: Initial PCB Layout

The class PCB layout was more compact than mine, so I switched to the class layout before starting PCB routing:

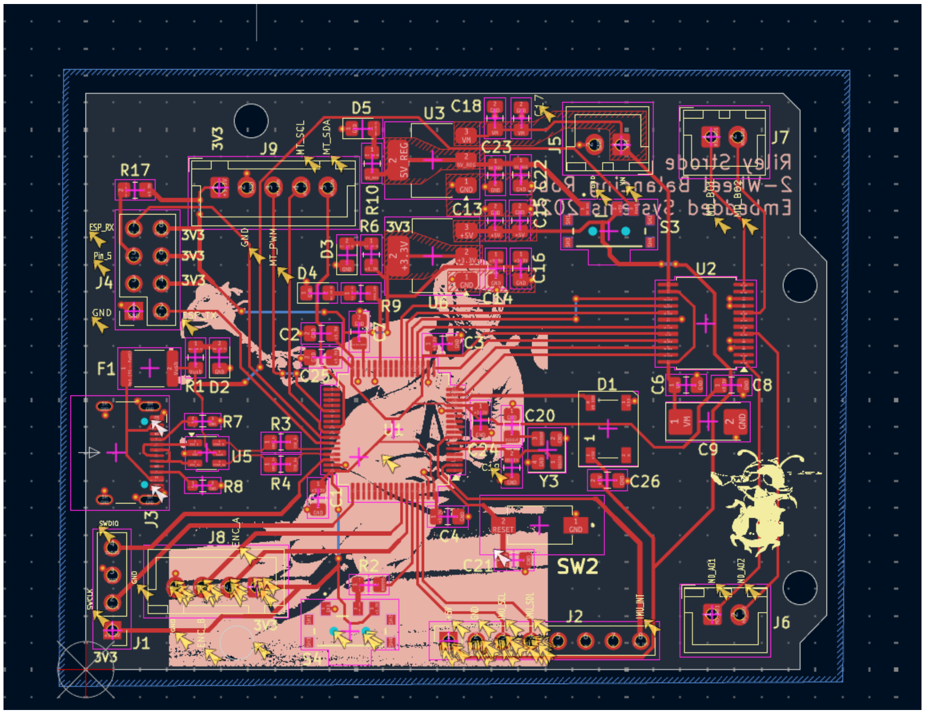

Figure 6: Class PCB Layout with My Routing

After routing all the components, I made sure to put a silkscreen of Lyle Thompson on the back because he has amazing balance (and he’s my favorite athlete).

Assembly Stage:

After ordering the boards from PCBWay, I soldered components to the board

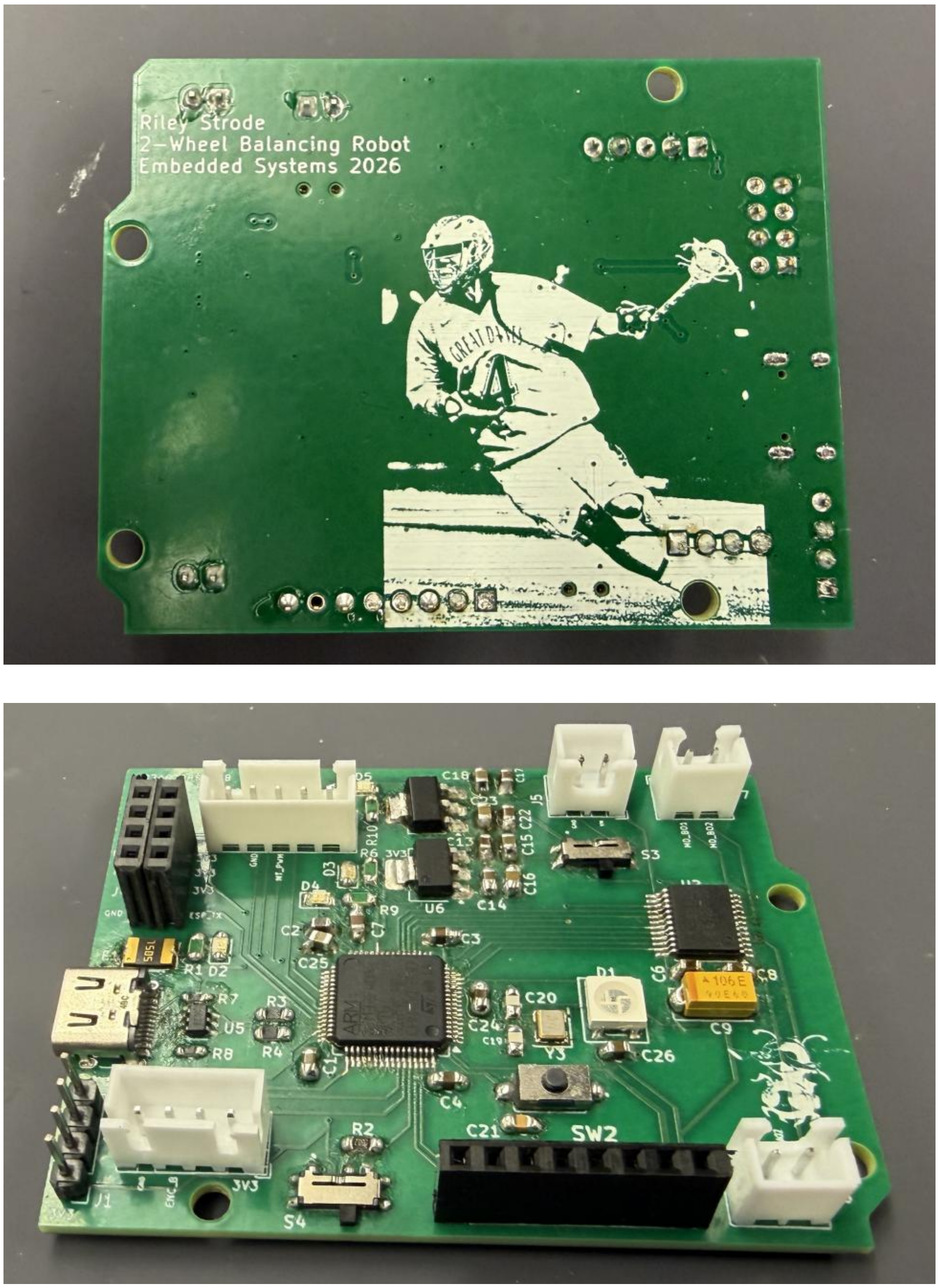

Figure 7: Soldered PCB Back & Front

Next, I designed a chassis in OnShape and cut it out of acrylic:

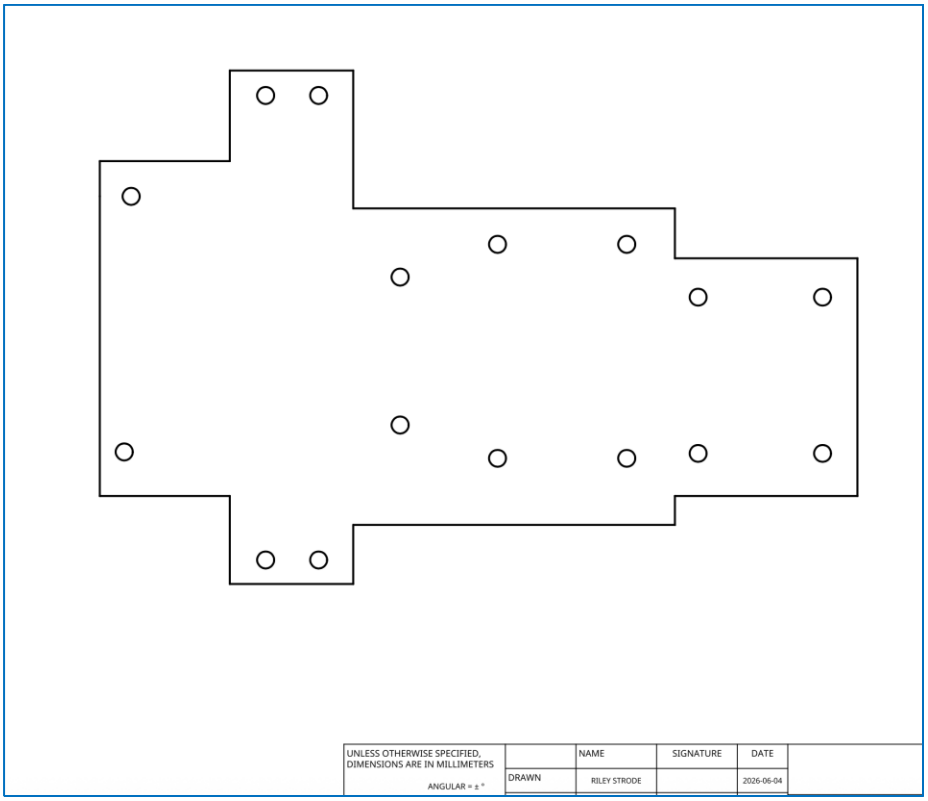

Figure 8: Chassis Design

Lastly, I attached the motors, wheels, PCB, and battery pack to the chassis:

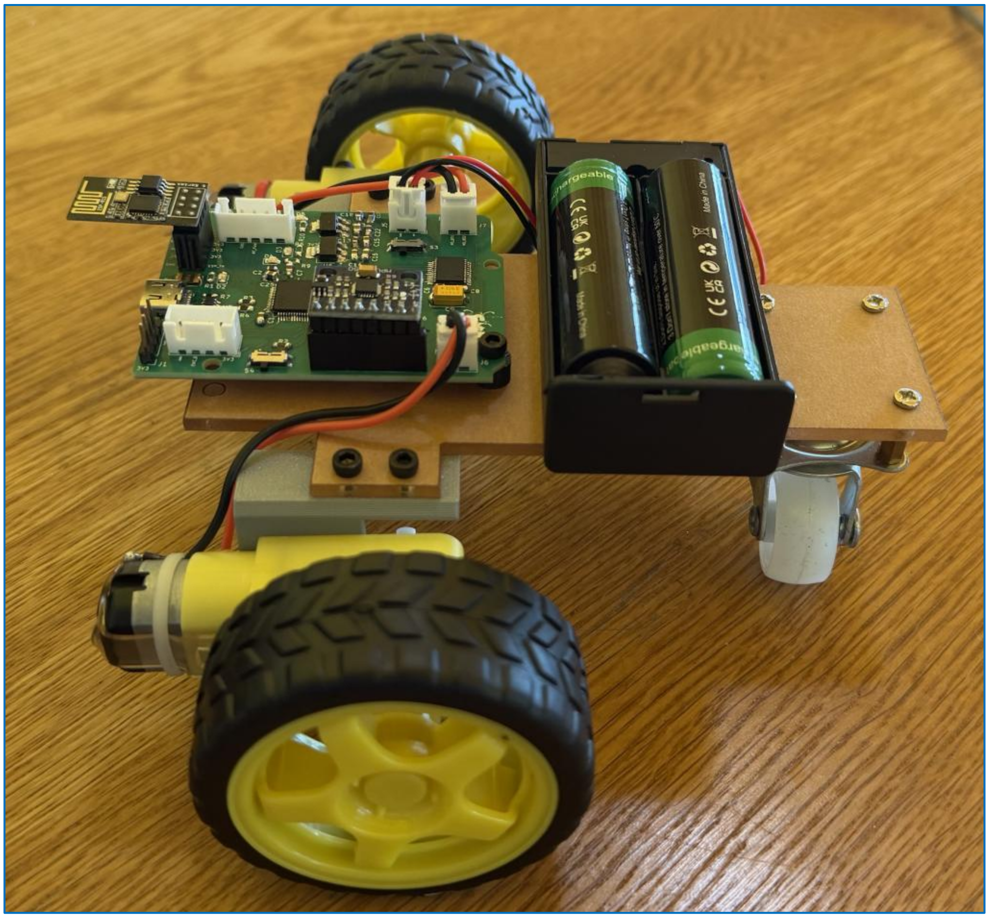

Figure 9: Fully Assembled Robot

Programming:

Using STM32CubeIde and STM32CubeProgrammer, I uploaded the code found at this link to the robot: https://github.com/riley-strode/ENCE_3231_Riley_Strode_2026/blob/main/Phase_D/2026_STM32F401_3Wheel_Balance_noEncoder_v2%20(1).zip

This code allows a user to control the robot remotely using the Wi-Fi module. By putting "192.168.4.1" into a web browser while connected to the robot's Wi-Fi module, the robot will respond to user inputs.

To see a video of my robot, view this link: https://github.com/riley-strode/ENCE_3231_Riley_Strode_2026/blob/main/Phase_D/IMG_3652.MOV

Bug Fixing:

The left motor (motor B) initially was not functioning at all. To try to figure out why, I probed my PCB (routing at motor driver chip), the motors, and the connectors with a multimeter and found no open circuits that would cause this lack of output.

After trying different codes, I found that the left motor responded only to high voltage outputs. My current hypothesis is that I partially damaged the motor driver chip (U2) while soldering or testing the robot since none of the routing connections are disrupted.